Literature resolution of the atomic JBW*-triple Tingley question in arXiv:1701.02916

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Classification

Status: literature already answered. The question isolated below was answered affirmatively by a separate paper of the same authors, posted one week after the source preprint. This note records the implication and makes no claim of a new proof.

Source question

Fernández-Polo and Peralta, *On the extension of isometries between the unit spheres of a C*-algebra and $B(H)$*, arXiv:1701.02916, prove that every surjective isometry

$$f : \mathbb{S}(B(K)) \longrightarrow \mathbb{S}(B(H))$$

extends to a surjective complex-linear or conjugate-linear isometry. They also obtain the analogous conclusion for ℓ_∞-sums of $B(H)$ spaces. At the end of the introduction they identify the next natural question: whether Tingley's extension problem has a positive answer for Cartan factors and atomic JBW*-triples.

Here Tingley's problem asks whether a surjective isometry between unit spheres extends to a surjective real-linear isometry of the ambient Banach spaces.

Answering paper

Fernández-Polo and Peralta, *Tingley's problem through the facial structure of an atomic JBW*-triple*, arXiv:1701.05112, was posted on 18 January 2017, seven days after arXiv:1701.02916. Its abstract already announces the required statement, and its main result (Theorem 2.9 in the published numbering) is:

> If E and B are atomic JBW*-triples and $f : \mathbb{S}(E) \to \mathbb{S}(B)$ is a surjective isometry, then there is a unique real-linear isometry $T : E \to B$ whose restriction to $\mathbb{S}(E)$ is f.

Surjectivity of T follows immediately from $T(\mathbb{S}(E)) = f(\mathbb{S}(E)) = \mathbb{S}(B)$. Thus the result is precisely an affirmative answer for arbitrary pairs of atomic JBW*-triples.

Every Cartan factor is an atomic JBW*-triple (and an arbitrary atomic JBW*-triple is an ℓ_∞-sum of Cartan factors), so the theorem also covers the Cartan-factor part of the source question.

How the later result connects to the source

The connection is explicit rather than inferred only from terminology:

1. The answering paper cites arXiv:1701.02916 as the recent $B(H)$ result.
2. Its introduction says that its purpose is to extend that result to atomic JBW*-triples.
3. Its theorem has exactly the sphere-isometry hypothesis and real-linear extension conclusion requested by the source question, for the full stated class.

The later paper was published as *J. Math. Anal. Appl.* **455** (2017), 750–760. Its DOI is `10.1016/j.jmaa.2017.06.002`.

Scope and novelty

This packet classifies only the source's stated Cartan-factor/atomic-JBW* question. It does not classify the unrestricted Tingley problem for arbitrary Banach spaces, and it does not rebrand the later authors' theorem as a new result. The resolution is unusually direct: same authors, a separate paper, an explicit continuation statement, and a theorem covering the entire source class.

Primary sources included

- `source_paper.pdf`: arXiv:1701.02916.
- `supporting_paper_1701.05112.pdf`: arXiv:1701.05112.